	with an increasing pitch/vertical separation and moving out of the plane of the paper.	B1
6a	The changing/alternating current in coil C produces a changing/alternating magnetic flux density (*B*) (since magnetic flux density is directly proportional to the current). Hence, coil D experiences a changing magnetic flux linkage since the direction of the B field is non-parallel to the plane/area of coil D. By Faraday's law, an e.m.f. is induced in coil D.	B1 B1 A0
6b	t = 5.0 ms, 15.0 ms or 25.0 ms	A1
6c	Since the gradient of the graph/rate of change of current at this point is the largest, (it corresponds to the greatest rate of change magnetic flux density/magnetic flux linkage.) Hence, induced e.m.f. is a maximum.	A1
6di	change in magnetic flux density = $B_f - B_i$ $= 1.30 \times 10^{-3} (I_f - I_i)$ $= 1.30 \times 10^{-3} (0 - (-2.4))$ $= 3.12 \times 10^{-3}$ $= 3.1 \times 10^{-3}$ T	M1

	$average\ \varepsilon = \left\|\frac{\Delta N\phi}{\Delta t}\right\| = \left\|\frac{N(\Delta B)A}{\Delta t}\right\|$ $= \frac{50(3.1\times10^{-3})(1.05\times10^{-3})}{5.0\times10^{-3}}$ $= 0.032\ \text{V}$	 C1 A1
6dii	From t = 10 ms to 15 ms, current in coil C is decreasing. The (magnitude of) magnetic flux linkage through coil D is decreasing. To oppose the decrease in flux, coil D induces a current that produces a magnetic field in the same direction as the magnetic field of coil C. Hence, the current in coil D flows in the same direction as coil C.	M1 M1 A0
7ai	$\Delta N / \Delta T$	B1